

Class
10

Concept of Information Technology-II

Class
10

Concept of Information Technology-II

Concept of Information Technology-II

Class X



BOARD OF SECONDARY EDUCATION, RAJASTHAN, AJMER

Text Book Writing Committee

Concept of Information Technology-II

Class X

Convenor

Atul Chaudhary

**Assistant Professor, Government Engineering College
Ajmer, Rajasthan**

Authors:

Dr. Anil Kumar Dubey

Lecturer

**Government Engineering College
Ajmer, Rajasthan**

Vinesh Jain

Assistant Professor

**Government Engineering College
Ajmer, Rajasthan**

Dinesh Khunteta

Assistant Professor

**Government Engineering College
Ajmer, Rajasthan**

Pramod Kaler

Lecturer

**Government Polytechnic College
Sikar, Rajasthan**

Text Book Syllabus Committee

Concept of Information Technology-II

Class X

Convenor

Dr. Vishnu Goel

**Director, Centre for E-Governance
Govt. Khetan Polytechnic College, Jaipur**

Members:

- 1. Dr. Anil Gupta, Assistant Professor
Computer Science & Engineering Department,
M.B.M. Engineering College, Jodhpur**
- 2. Shri Harji Ram Choudhary, Assistant Professor
Govt. Engineering College, Ajmer**
- 3. Shri Dalpat Singh Songara, Assistant Professor
Govt. Girl's Engineering College, Ajmer**
- 4. Shri Amarjeet Punia, Assistant Professor
Govt. Girl's Engineering College, Ajmer**
- 5. Shri Vishnu Prakash Sharma, Assistant Professor
Govt. Engineering College, Ajmer**
- 6. Shri Rajesh Kumar Tiwari, Principal
Govt. Sr. Sec. School Jotaya, Sarwar, Ajmer**

Preface

This book covers essential concept of Information Technology. The overall aim of this book is to introduce you to various advance processing tools, cyber crime, e-commerce, application softwares, system softwares, internet and its applications.

The coverage of this book is according to the recently revised syllabus of Board of Secondary Education, Rajasthan. This book comprises of 14 chapters and each chapter has its own significance.

Chapter 1 to 5, these chapters are part of first unit and cover advance processing tools which includes MS Excel, MS PowerPoint and MS Picture Manager.

Chapter 6 to 9, these are based on cyber crime and e-commerce, which includes introduction to cyber crime & cyber law, e-commerce/e-business, e-business security, privacy and legal requirements, e-commerce payment system and these are part of unit second.

Chapter 10, this chapter discusses application and system software which includes proprietary software, open source software, operating system and is part of unit third.

Chapter 11 to 14, these chapters are part of fourth unit and cover internet and its application, computer network, web, electronic mail, virus and antivirus.

Our thanks to all, who supported us directly or indirectly collect the data. Most importantly, our thanks to our family members, because without their support this book was a dream which would not have been turned into reality.

Suggestions for improvement in the book are welcome.

Authors

SYLLABUS

UNIT I: ADVANCE PROCESSING TOOLS

Introduction to MS-Excel, Comparison between MS-Word and MS-Excel, Worksheets and Workbooks: Creating a Workbook, Opening, Labeling, Format Workbook Tabs, Reposition Sheets, Naming, Adding, Deleting, Hiding, Unhiding, Saving Workbooks and Worksheets. Navigating MS Excel, Insert Cells, Rows and Columns, Delete Cells, Rows or Columns, Merge, Splitting, Hiding Columns and Rows, Unhiding Column and Rows, Format, Filter and Sort of Cells, Headers and Footers, Set Margins For Headers and Footers, Information About Printing: Select Print Area, Print a Range of Pages. About Entering Information into Excel: Entering Data, Entering Labels, Entering Values, Multiple Entries, Copying & Pasting of Cells, Rows and Columns. Filling Cells with a Series of Data, Editing Cell Data, Find and Replace, Go To Cell Data, Locking Rows and Columns by Splitting Panes and Freezing Panes, Spell Check, Autocorrect, Track Changes, Accept and Reject Changes, Comments. Formatting A Worksheet: Format Painter, Font Styles, Font Size, Adding Border and Colours to Cell, Changing Rows and Column width, Changing Rows and Column width Using the Mouse, Applying Number Formats, Creating Custom Number Formats, Align Cell Contents, Cell Styles, Creating your own Cell Styles, Conditional Formatting. Adding Elements to A Workbook: Adding Images, Modifying Image, Charts: Types of Charts, Chart Tools, Creating Charts, Modifying Charts, Moving Charts, Organizational Charts, Spark Line, Formulas and Calculations: Definition and Explanation of Formulas and Calculations, Mathematical Operators, Creating a Formulas, Creating Functions, References, Excel Forms, Tables, Creating a Table, Inserting Rows and Columns Into a Table, Data Validation, Finding Invalid Entries and Auditing, Page Margins, Page Orientation, Page Breaks, Sharing Worksheets and Workbooks, Importing and Exporting Data.

MS Power Point: Opening, Opening an Existing Presentation, Saving and Closing a Presentation, Changing Views, Creating a New Presentation, Adding a Slide, Changing a Slide Layout, Entering Text on a Slide, Changing Text Formats, Using The Format Painter, Bullets, Alignments Text, Using Templates, Adding a Picture, Using The Slide Master, Adding Headers, Footers and Speaker Notes, Arranging Slides, Introduction to Drawing Tools, Inserting and Formatting Picture Files, Insert a Table, Chart, Smart Art, Hyperlink, Transition Effects, Animation Effects, Sound Clip, Running a Slide Show, Creating a Custom Show.

MS Picture Manager: Open a Picture, Find the properties of picture, Auto Correct, Brightness and Contrast Enhancement, Color Enhancement, Crop Settings, Rotate and Flip Settings, Resize Settings, Compress Settings.

UNIT II: CYBER CRIME AND E-COMMERCE

Cyber Law, Technical Aspects of Cyber Crime, Computer Viruses, Social Engineering, Phishing, Software Piracy, Intellectual Property, Mail Bombs.

E-Commerce: features, Advantages to Organizations, Advantages to Customers, Advantages to Society, Technical Disadvantages, Non-Technical Disadvantages, E-Commerce Business Models, The six components of a marketing plan for small business, Perform SWOT Analysis, Business Philosophy and Vision, E-Business Security, Security, Types of Security Risks Encountered on an Intranet and Extranet, Firewalls and their Evolution, Types of Firewall,

Common Firewall Filtering Techniques, Cryptography, Digital Signature, Virtual Private Network (VPN), Types of VPN, Measures to Ensure Security, Modes of Electronic Payments, Third-Party Payment Processor, Payment Gateway, Traditional Marketing, Examples of Traditional Advertising, Internet Marketing, Protection of Privacy and Intellectual Property

UNIT III: APPLICATION AND SYSTEM SOFTWARE

Definition of Computer Data, Information, Computer Instruction, Computer Program, Software: Application and System Software, Uses and Examples of Application Software and System Software, Proprietary Software and Open Source Software (OSS), Foundation of the Open Source Software, Comparison Between OSS and Proprietary Software, Reasons for Adoption of Open Source Software, Advantages and Disadvantages of OSS, Operating System: Microsoft Window, Linux, Open Office

UNIT IV: INTERNET AND ITS APPLICATION

Objectives of Computer Network, Components of Computer Network, Advantages and Disadvantages of Computer Network, Application of Networks, Local Area Network, Wide Area Network, Internet, Future of the Internet, Applications of Internet, Different ways to access the Internet, Services on Internet, Communication on Internet, Internet Protocol, HTTP, FTP, SMTP, Telnet, Intranet, WWW (World Wide Web), Websites, Web Browsers, Internet Explorer, The URL address, Surfing the Internet, Chatting on Internet, Conferencing on Internet, E-Mail or Electronic Mail, Types of E-Mail Services, Uses of Mail Services, Basic Issues of Email, Virus, Anti-Virus, Virus Protection Software, Protection of the Computer from Virus, Updating the software, Type of Internet access, Online Services, Internet Services Provider, Hypertext and Hyperlinks, Favorites or Bookmarks, Cookies, Bluetooth, Wi-Fi, DHCP, Proxy Setting, IP address, Subnet Mask, Gateways, DNS

Chapter Index

Unit No.	Chapter No.	Title	Page No.
Unit I			
	Chapter 1	Introduction to MS Excel	1 - 45
	Chapter 2	Formatting of Worksheet	46 - 62
	Chapter 3	Adding Elements to a Workbook	63 - 96
	Chapter 4	MS PowerPoint	97 – 137
	Chapter 5	MS Picture Manager	138 - 151
Unit II			
	Chapter 6	Introduction to Cyber Crime & Cyber Law	152-162
	Chapter 7	E-Commerce/E-Business	163-174
	Chapter 8	E-Business Security, Privacy, and Legal Requirements	175-184
	Chapter 9	E-Commerce Payment System	185-192
Unit III			
	Chapter 10	Application and System Software	193-212
Unit IV			
	Chapter 11	Computer Network	213-226
	Chapter 12	Web & Electronic Mail	227-241
	Chapter 13	Virus And Antivirus	242-246
	Chapter 14	Internet	247-258